C U R R I C U L U M   V I T A E

Rahul Rawat

Technology, Data and Innovation Intern

PERSONAL DETAILS

Address	C2 16, 68159 Mannheim, Germany
Phone	015563603340
Email	rahulrawat2r@gmail.com
LinkedIn	linkedin.com/in/rahulrawat2r
GitHub	github.com/rahulrawat20022002
Portfolio	rah-portfolio.pages.dev
Date of birth	20 February 2002
Nationality	Indian, student visa with valid work permit
Availability	Werkstudent 20 hours per week immediately, full time from April 2027

PROFILE

Data Science and Analytics Master student at SRH Heidelberg based in Mannheim, with hands on experience in Python, PySpark, SQL, and cloud data platforms on real production style data. I have shipped a fairness by design credit scoring system built with mitigation techniques from IBM AIF360, a real time flight tracking pipeline processing over **128 thousand** records with PySpark on Google Cloud, and an automated Bronze to Silver to Gold BigQuery medallion architecture with a leakage free BigQuery ML classifier and a five page Looker Studio dashboard. Comfortable translating business requirements into data driven insights, building interactive analytics for non technical stakeholders, and operating inside regulated environments, I am the right fit for the Technology, Data and Innovation internship at Deutsche Bank in Frankfurt.

PROFESSIONAL EXPERIENCE

Oct 2025 to Mar 2026
Heidelberg

eRay GmbH

Data Scientist

- Built an end to end recursive time series pipeline forecasting chlorophyll a, turbidity, pH, and dissolved oxygen for a German lake, delivered as a six month collaboration with SRH University Heidelberg.
- Benchmarked six models head to head, Ridge, Gradient Boosting, LightGBM, XGBoost, CatBoost, and Prophet, and used CatBoost multi quantile regression to produce asymmetric **80 percent** prediction intervals for decision support under uncertainty.
- Enforced strict anti leakage rules across the pipeline, surfacing the honest finding that pH and dissolved oxygen are physically predictable while chlorophyll a and turbidity are not without live optical sensors.
- Reconstructed missing winter readings with MICE imputation and engineered a synthetic winter decay forecast canvas so tree based models stopped flatlining during recursive prediction.
- Wrapped the whole pipeline in an orchestrator with gate checks and velocity and ecological bounds that halts on failed imputation rather than letting bad data cascade downstream.

Technologies: *Python, CatBoost, Prophet, scikit learn, MICE*

PERSONAL PROJECTS

2025

CreditIQ: Fairness by Design Credit Scoring

- Raised the model's Disparate Impact ratio from a failing **0.79** to a compliant **0.88**, measured against the EU AI Act and AGG **80 percent** fairness threshold, using AIF360 mitigation and threshold calibration on a real credit dataset.
- Diagnosed a hidden intersectional bias where younger women were still penalised even after single axis age bias was fixed, measured through SHAP driven subgroup analysis, then designed a four way age by gender threshold matrix to correct it without over correcting into reverse discrimination.
- Cut the false negative rate from **44 percent** to **16.7 percent** while holding accuracy at **75 percent**, measured on the held out test split, by documenting the fairness accuracy trade off as a deliberate and regulator defensible decision rather than a hidden compromise.
- Shipped a Streamlit decision support tool that gives finance managers a recommendation plus a plain language LLM generated explanation, measured against GDPR Article 22 and EU AI Act Article 14 human in the loop requirements, and backed the pipeline with unit tests at **100 percent** branch coverage and a full regulatory write up.

Technologies: *Python, scikit learn, AIF360, SHAP, Streamlit*

2025

Real Time Flight Tracking Data Pipeline

- Collected live flight positions from the OpenSky Network API every **30 seconds** and enriched them against airport, aircraft, and weather data across four sources, measured by a clean joined table covering more than **128 thousand** records, using Python collectors and PySpark cleaning on Google Cloud.
- Shaped raw collector output into analysis ready tables with dbt and computed each aircraft's nearest airport along the way, measured by consistent nearest airport labels across the historical dataset, using PySpark for heavy lift and dbt for the modelling layer.
- Orchestrated the whole system with Apache Airflow so batch and real time layers refresh automatically every **15 minutes**, measured by unattended DAG runs on GCS backed storage and Dataproc compute.
- Surfaced findings in a Tableau workbook backed by Python statistics through TabPy, measured by the finding that air traffic drops **4.4 times** in heavy rain and clusters around hubs like Frankfurt and Munich, using dbt aggregates as the visual layer feed.

Technologies: *Python, PySpark, BigQuery, dbt, Apache Airflow, GCP Dataproc and GCS, Tableau with TabPy*

2025

Movie Analytics and ML Pipeline on GCP

- Built an end to end batch pipeline pulling movie data from a public API into a GCS data lake and processing it through Bronze to Silver to Gold medallion architecture in BigQuery, measured by a fully automated Cloud Scheduler trigger with no manual intervention required, using Cloud Run for compute.
- Engineered the Silver layer for trustworthy analytics with schema enforcement, safe type casting, deduplication via window functions, and genre normalisation into a relational model, measured by clean referential integrity across all downstream Gold tables.
- Trained a BigQuery ML classifier that predicts whether a film will be a hit before release, measured by leakage free evaluation, by deliberately splitting features into two tables so only pre release signals feed the model.
- Built Gold layer aggregates and a five page Looker Studio dashboard answering concrete business questions on genre ROI, foreign language growth, and release season timing, measured against the ML early warning view, and secured the system with a least privilege service account and Secret Manager.

Technologies: *GCP BigQuery, Cloud Run, GCS, Cloud Scheduler, BigQuery ML, Python, SQL, Looker Studio*

EDUCATION

Apr 2025 to Present
Heidelberg

M.Sc. Data Science and Analytics
SRH University of Applied Sciences Heidelberg, GPA 1.9

2019 to 2023
Greater Noida, India

Bachelor of Technology in Computer Science
GL Bajaj Institute of Technology and Management, CGPA 7.3 of 10

RESEARCH AND THESIS

2022 to 2023
Greater Noida, India

Bachelor Thesis, Diabetes Prediction Using Machine Learning
GL Bajaj Institute of Technology and Management, IEEE style paper

- Built a full end to end machine learning pipeline comparing six classifiers on a clinical dataset of **768 patients**, using **10 fold** cross validation and per model confusion matrices to make the model comparison auditable.
- Caught biologically impossible zero values that the original authors had overlooked, then applied IQR based outlier removal and proper imputation to restore data integrity before any model fit.
- Chose ROC AUC over accuracy as the headline metric for a **65 to 35** imbalanced dataset, avoiding a misleadingly rosy accuracy figure that would have hidden real error patterns in the minority class.
- Wrote up findings as an IEEE style paper including an honest section on limitations and what to do differently in a follow up study, publishable in substance.

Technologies: *Python, scikit learn, Pandas, Seaborn*

CERTIFICATIONS

- AWS Academy Graduate: AWS Academy Cloud Foundations, issued July 2025.
- Google Data Analytics: Foundations, Data, Data, Everywhere, issued April 2025 on Coursera.
- SAS Certified Specialist: Visual Business Analytics Using SAS Viya, issued May 2025.

ACHIEVEMENTS

- USAII Global AI Hackathon 2026: Finalist at Graduate Level, awarded by the United States Artificial Intelligence Institute for innovation, technical creativity, and applied AI on real world challenges.

LANGUAGES

English	Fluent, written and spoken
German	B1 in progress toward B2
Hindi	Native

Mannheim, 21 July 2026
Rahul Rawat